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**B.Sc. (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER
EXAMINATION, 2023**

DIGITAL SYSTEMS DESIGN

[According to the New Syllabus]

Subject Code: 510221

Examination Code : 5612

Time—3 hours

Full marks—80

4 2 1

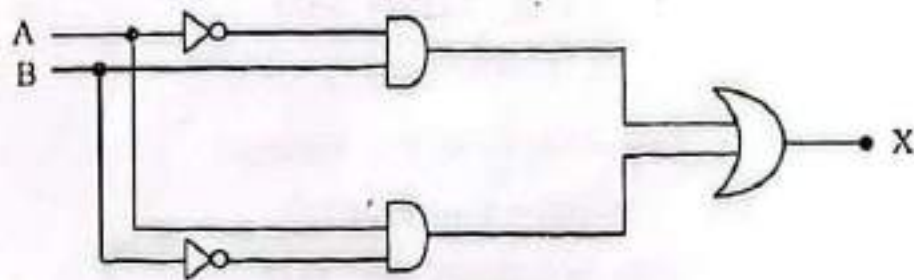
[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|---|-------|
| 1. (a) What is analog and digital system? | 2 |
| (b) Write down the advantages of digital techniques. | 4 |
| (c) Convert the following numbers into given base numbers: | 6 |
| (i) $(562.65)_{10} = (?)_2 = (?)_8$ | |
| (ii) $(BCD)_{16} = (?)_8$ | |
| (d) Define the following terms with example : | 8 |
| (i) Hexadecimal Code | |
| (ii) BCD | |
| (iii) Unicode | |
| (iv) Excess 3 Code. | |
| 2. (a) Design a logic circuit with inputs P, Q and R. The output will be high whenever $P = 0$ or $Q = R = 1$. | 6 |
| (b) List the applications of Encoder and Decoder. | 4 |
| (c) What is Encoder? Design an octal to binary encoder. | 5 |
| (d) What do you mean by <u>shift register</u> ? Mention <u>different types of</u> shift register. | 1+4=5 |

[Please turn over

Marks

- (a) Define min term and max term. Prove that $X+X=X$ and $X \cdot X=X$. 2+4=6
(b) Determine the output expression for the following circuit: 4



- (c) What is karnaugh map? Minimize the following logic functions and realize using NOR gates: 1+5=6

$$F(A, B, C, D) = \pi M(1, 2, 3, 8, 9, 10, 11, 14), d(7, 15)$$

- (d) Explain with proper diagram of a clocked S—R flip-flop. 4

- (a) What is binary parallel adder? Describe a 4-bit parallel adder with neat sketch. 1+5=6

- (b) Draw the logic circuit of latch and truth table using NAND gates. 4

- (c) Reduce the following function using K-map: 5

$$F = \bar{C}(\bar{A}\bar{B}\bar{D} + D) + A\bar{B}C + \bar{D}$$

- (d) Explain parity generator and checker with suitable diagram. 5

- (a) What is subtractor? Describe full subtractor circuit with logic diagram and truth table. 7

- (b) Define ripple counter. Design a 4-bit ripple counter and sketch its transitional diagram. 8

- (c) Design a J-K flip-flop and describe its functions using truth table. 5

6. (a) Write down the function of transducer. 3

- (b) What is DAC? Explain the operation of DAC with proper diagram. 1+6=7

- (c) A certain 12-bit BCD digital to analog converter has a full scale output of 9.99V. Determine— 6

(i) Percentage resolution (ii) The converter's step size.

- (d) What are the advantages of R/2R ladder DAC over binary weighted registers. 4

**B.Sc. (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER
EXAMINATION, 2021**

DIGITAL SYSTEMS DESIGN

[According to the New Syllabus]

Subject Code : 510221

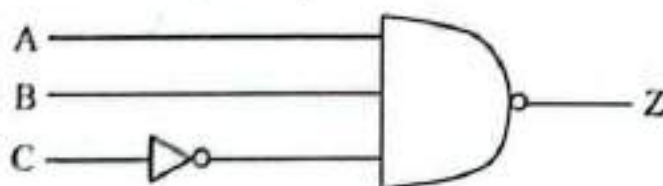
Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions. Different part of a questions must be answered sequentially.]

- | | Marks |
|--|-------|
| 1. (a) What is analog and digital system? Write down the advantages of digital techniques. | 5 |
| (b) Convert the following numbers into given base numbers :
(i) $(110101)_2 = (?)_{10}$
(ii) $(129A-B8C)_{16} = (?)_2$
(iii) $(225-225)_{10} = (?)_2$ | 6 |
| (c) Explain the method of subtraction using 2's complement method with example. | 5 |
| (d) Write short notes about the following terms :
(i) ASCII Code
(ii) Gray Code. | 4 |
| 2. (a) Define universal gate. Prove the universality of NAND gate and NOR gate. | 6 |
| (b) State and prove De-Morgan's theorem. | 5 |
| (c) Determine the output expression for the circuit of the following figure and simplify it using De-Morgan's theorem. | 4 |



- (d) Simplify the expression.

$$Z = ABC + AB\bar{C} + (\bar{A} \cdot \bar{C})$$

[Please turn over]

	Marks
3. (a) What is Flip-Flop? Write down the applications of Flip-Flop.	3
(b) Explain a clocked J-K Flip-Flop.	6
(c) Define NOR gate latch? Explain clocked S-C Flip-Flop.	6
(d) Explain how J-K Flip-Flop can be converted to :	5
(i) D-type Flip-Flop	
(ii) T-type Flip-Flop.	
4. (a) To construct a Full Adder circuit by using Half-Adder circuit.	6
(b) What is Encoder? Design a Octal to Binary Encoder?	6
(c) What is ripple counter? Explain a 4-bit ripple counter with diagram.	5
(d) Write down the difference between synchronous and asynchronous.	3
5. (a) What is Demultiplexer? Sketch the logic diagram with truth table of 1 to 8 line Demultiplexer.	6
(b) What is NAND latch? Describe S-R Flip-Flop with logic diagram..	6
(c) Draw and explain 5-bit shift register.	5
(d) Mention the major drawbacks of S-R Flip-Flop.	3
6. (a) Outline the steps that take place when the CPU reads from memory.	5
(b) Differentiate between EPROM and EEPROM.	5
(c) Describe the basic operation of the digital ramp ADC.	6
(d) What is the major advantages of SAC over a digital ramp ADC?	4

$A \oplus B$
 00
 01
 10
 11
 01
 10
 00
 11

$$S = A \oplus B$$

$$C = A \cdot B$$



000
 001
 010
 011
 100
 101
 110
 111

DIGITAL SYSTEMS DESIGN

[According to the New, Syllabus]

Subject Code : 510221

Examination Code : 5612

Time-3 hours

Full marks 80

1. (a) Describe the main functional units, of a digital computer. 5
 (b) Write short notes about the following terms: 4
 (i) BCD code
 (ii) Unicode
 (c) Represent the decimal number 396 in
 (i) Binary form
 (ii) Gray Code
 (iii) Excess-3 code
 (d) What are the steps actually required for processing an analog signal using digital hardware? Explain briefly.
2. (a) Implement the logic circuit that has the expression $X = AB(C + D)$ 6
 using only NOR and NAND gates.
 (b) Differentiate between binary and BCD code. 3
 (c) What is Boolean algebra? Explain. the Boolean function $F = A + BC$ in a sum of minterms. 6
 (d) Design a logic circuit with inputs L, M and N. So that the output P is high. Whenever the majority of inputs are high. 5
3. (a) What is setup and hold times? Explain J-k flip-flop with suitable diagrams. 8
 (b) How can a J-k flip-flop be modified to operate as a D flip-flop and a T flip-flop? 6
 (c) What is NAND gate latch? Explain clocked S-C flip-flop with timing diagram. 2+4=6
4. (a) Explain the difference between a DEMUX and a MUX. 4
 (b) Design and explain four-input multiplexer. 6
 (c) What is semiconductor memory? Discuss the classification of semiconductor memory. 6
 (d) Describe static and dynamic RAM. 4
5. (a) What is Decoder? Explain 3 to 8 line Decoder with truth table and circuit diagram. 6
 (b) What is Karnaugh map? Using K-map simplify the following expression: 4
 $F(A, B, C, D) = \sum m(1, 3, 4, 6, 8, 9, 11, 13, 15) + d(0, 2, 14)$
 (c) Show how to wire a 74LS293 as a MOD-14 counter. 4
 (d) Design a full adder circuit using basic logic gates and explain its operations with truth table. 6
6. (a) State the purpose of analog to digital conversion. 5

- | | |
|--|---|
| (b) Mention some applications of Decoder. | 5 |
| (c) Explain the operation of a 4 bit R-2R ladder DAC. | 6 |
| (d) Determine the output expression for the following circuit: | 4 |

**B.Sc. (HONS.) IN COMPUTER SCIENCE AND ENGINEERING
FIRST YEAR SECOND SEMESTER EXAMINATION, 2019**

(According to the New Syllabus)

CSE=510221

(Digital Systems Design)

1. (a) Define digital system. Write down importance of digital system.
(b) Discuss the major parts of a digital computer with required diagram.
(c) Define ASCII and Gray Code. Convert $(123.45)_{10} = (?)_2 = (?)_{Hex} = (?)_{BCD}$
(d) Write down Demorgan's theorem. Prove Demorgan's theorem with 3 variables.
2. (a) What a universal gate? Prove the universality of NOR gate.
(b) Define: K-map. Using K-map simplify the expression $Y = c(ABD + D) + ABC + D$, also draw the logic diagram of simplified Y.
(c) That is don't care condition of K-map? Explain.
(d) Consider the following Boolean logic diagram. If you want to get the output value $x = 0$ and $x = 1$, then what will be the input values of A, B, C, D:
4. (a) Define flipflop. Draw the logic circuit-of an S-R flipflop with HAND latch and explain its logic operation
(b) Design and discuss the MOD-6 counter.
(c) Write down the truth table for a full adder. Draw the logic circuit and explain its operation
5. (a) Explain the operation of successive approximation ADC with required diagram.
(b) What is the largest value of output voltage from an eight-bit. DAC that produces: 1 -OV for a digital input of 00101000?
(c) Differentiate between EPROM and DRAM.
(d) Mention the advantageous of successive approximation ADC 3 over digital Ramp ADC.
6. (a) What is encoder? Explain the working principle of 8 line to 3-line encoder with logic diagram and truth table.
(b) Define MUX and DMUX. Why MUX is called data selector? and explain it.
(c) Draw and explain BCD to 7 segment decoders.
7. (a) What are the differences between combinational logic circuit and sequential logic circuit?
(b) Define memory. Discuss the basic organization of a memory unit of 3×4 bits.
(c) How many 32k x8 RAM chips are needed to provide a memory capacity of 256k bytes?
(d) Describe the functional parts of an ALU.

B.Sc (HONS) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2018

[According to the new Syllabus]

CSE-510208

Examination Code: 5612

(Digital System Design)

1. (a) Describe the organization of a digital computer with figure. 6
 (b) Define parity bit. Why it is significant for transmission of digital signals? 4
 (c) Using 2's complement system subtract 5 from 9. 4
 (d) Define the following terms with examples :
 (i) ASCII Code
 (ii) BCD Code
 (iii) Excess-3 Code
2. (a) Prove the universality of NAND gate. 4
 (b) Design a logic circuit with inputs P, Q and R so that output S is HIGH whenever P is 0 or whenever Q = R = 1. 6
 (c) Explain how a de-multiplexer can be used as a decoder. 4
 (d) Define : 6
 (i) Setup time
 (ii) Hold time
 (iii) Propagation delay
3. (a) Describe the basic DRAM cell with read, write and refresh operation. 4
 (b) Explain the differences between SRAM and DRAM. 5
 (c) Describe a basic PROM storage cell. 5
 (d) What is shift register? Explain 4 bit shift register with waveform. 5
4. (a) What is LCD? Explain BCD to 7 segment decoder. 6
 (b) Draw and explain a complete 4 bit parallel adder with register. 7
 (c) Design and construct a full adder circuit. Explain its operation with truth table. 7
5. (a) State and prove the De-Morgan theorem with the help of truth table for three variables. 4
 (b) What is Decoder? Explain 3 to 8 line. Decoder with truth table and circuit diagram. 6
 (c) Minimize the following function and realize using NAND gate only : 6

$$f = \sum m(1, 3, 5, 8, 9, 11, 15) + d(10, 13)$$

 (d) What should be done in BCD addition when the sum is greater than 9? Explain with example. 4 greater
6. (a) What is race-around condition? Explain how it can be eliminated. 5
 (b) Perform the following flip-flop conversions : -
 (i) S-R Flip Flop into D Flip Flop
 (ii) J-K Flip Flop into T Flip Flop
 (c) Describe the operation of a successive approximation ADC. 6
 What are the main advantages of this type of ADC? 5
 (d) What are the differences between combinational and